



Systems of Inequalities in Representational Measurement Theory: Foundations, Scope, and Formalization

1. Overview and Context

Determining the existence of solutions to systems of inequalities is a fundamental problem across mathematical optimization, real algebraic geometry, and variational analysis. In the foundational literature of mathematical psychology, decision theory, and the social sciences—exemplified by Dana Scott's seminal paper, "*Measurement Structures and Linear Inequalities*" (1964)—this problem takes center stage within **Representational (Axiomatic) Measurement Theory**.

Measurement theory investigates the formal conditions under which qualitative, empirical observations (such as preference orderings, perceived intensity, or comparative likelihoods) can be faithfully represented by real numbers.

2. Mathematical Foundations: Linear Inequalities and Representation Theorems

When qualitative comparisons on finite sets (e.g., $A \succsim B$) are translated into numerical relations (e.g., $u(A) \geq u(B)$ or $P(A) \geq P(B)$), the problem reduces to determining whether a finite system of linear inequalities has a non-trivial real solution.

Core Mathematical Machinery

1. Theorems of the Alternative (Convex Optimization & Polyhedral Geometry):

- **Farkas' Lemma & Motzkin's Transposition Theorem:** These results establish that a system of linear inequalities $Ax \leq b$ has a solution if and only if a dual system has no solution.

- In measurement theory, dual non-solvability yields finite **cancellation axioms** (such as the Kraft–Pratt–Seidenberg conditions for finite comparative probability), guaranteeing the existence of an agreeing numerical scale.

2. **Fourier–Motzkin Elimination:**

- An algorithmic technique for projecting systems of linear inequalities onto lower-dimensional subspaces, providing constructive proofs for feasibility.

3. **Model Theory and Relational Structures:**

- Establishing representation theorems as structure-preserving homomorphisms from an empirical relational structure $\langle A, R_1, \dots, R_k \rangle$ into a numerical structure $\langle \mathbb{R}, \geq, +, \dots \rangle$.

3. Disambiguation: Measurement Theory vs. Measure Theory

Although their names are frequently conflated, **Measurement Theory** and **Measure Theory** address fundamentally distinct questions:

Dimension	Measure Theory	Measurement Theory
Primary Domain	Real Analysis, Functional Analysis, Probability	Mathematical Psychology, Decision Theory, Foundations of Science
Core Question	How do we rigorously assign size, area, volume, and integration to abstract sets?	Under what axiomatic conditions can empirical relations be mapped homomorphically to numbers?
Key Objects	σ -algebras, Borel measures, L^p spaces, Lebesgue integration	Relational structures, scales (ordinal, interval, ratio), semiorders, conjoint structures
Foundational Figures	Lebesgue, Borel, Radon, Carathéodory	Scott, Suppes, Luce, Krantz, Tversky

Point of Contact: The two fields briefly intersect in **subjective probability**, where measurement theory formulates axioms ensuring that a comparative probability relation $\succsim$ induces an agreeing

numerical probability distribution—which is a measure in the analytic sense.

4. Formalization Landscape in Interactive Theorem Proving (Lean / Mathlib)

As of current developments in Lean 4's mathematical library (`Mathlib`):

- **Dedicated Measurement Theory:** There is currently no dedicated library for representational measurement theory, Scott's representation theorems, semiorders, or additive conjoint measurement.
 - **Underlying Building Blocks in Mathlib:**
 - **Convex Geometry & Duality:** `Mathlib.Analysis.Convex.Cone.InnerDual` contains geometric formulations of hyperplane separation theorems and dual cone properties (Farkas' Lemma).
 - **Inequality Decision Procedures:** `Mathlib.Tactic.Linarith` and the `omega` tactic internally implement Fourier–Motzkin elimination and Presburger arithmetic decision procedures to certify the solvability of linear inequality systems.
 - **Order Theory:** `Mathlib.Order.Extension.Linear` contains Szpilrajn's Extension Theorem, and `Mathlib.Order.Hom.*` provides general machinery for order embeddings and relational homomorphisms.
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